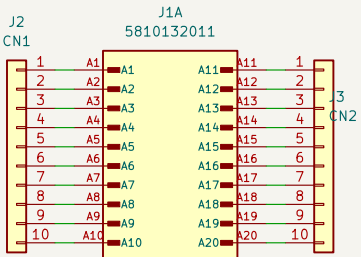
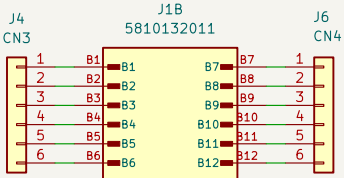


PCB/Enclosure
Output Connectors



Enclosure Header A (20-pin)
Peripheral Connector



Enclosure Header B (12-pin)
CAN-BUS + PWR Connector

N1
Cinch 5810130065 Enclosure



These are the main inputs/outputs from the enclosure that will connect to the wiring harness on the car.

Internally they are connected to pin headers inside the enclosure so that the pinout is completely adjustable for prototyping different pinouts and connections on the MCU

VCU Module TOP-LEVEL ARCHITECTURE

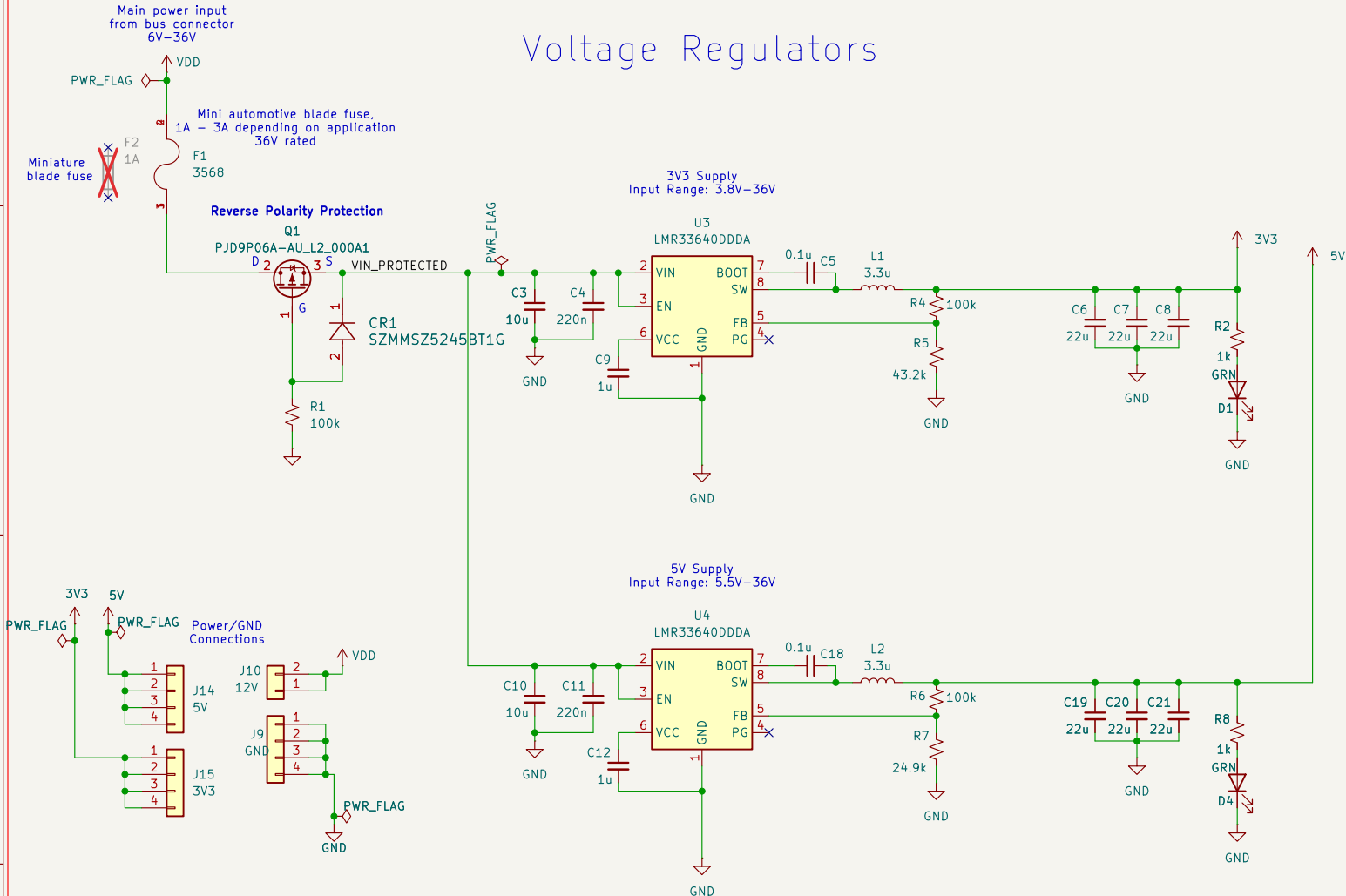
The "VMS VCU Dev Board" is the first iteration of the VCU module using a custom PCB designed to fit in a Cinch 2.25" x 3.35" enclosure

The board accepts up to a 36V DC power source (needs to be manually connected from bus connector) and contains both 3.3V and 5V regulators to power the MCU and any external sensors/components

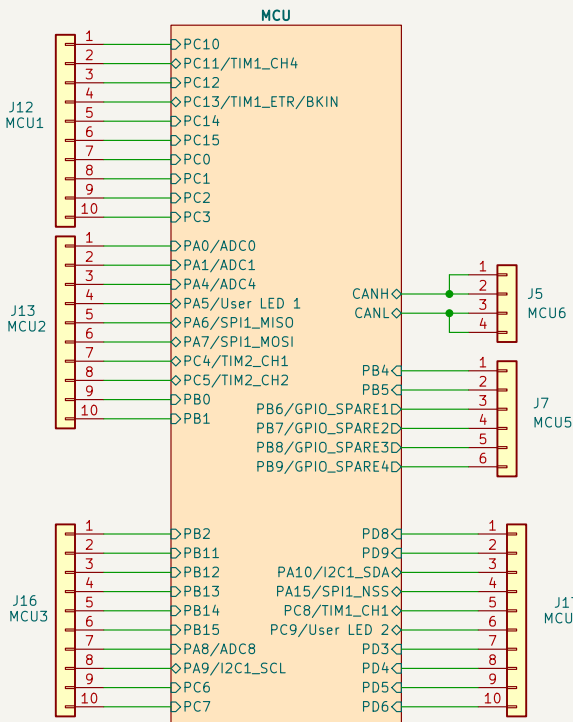
The Cinch enclosure header consists of a 12-pin connector which is to be designated as the main bus connection point and a 20-pin connector which is to be used to interface with external components and sensors

Pins from the PCB header will connect to internal headers that can be manually connected via jumpers to the appropriate source (MCU pin, 5V, 3.3V, CAN H/L, GND, etc.)

Voltage Regulators



MCU



File: MCU.kicad_sch

VIKING MOTOR SPORTS (VMS)

Sheet: /
File: VCU_Module.kicad_sch

Title: **VEHICLE CONTROL UNIT (VCU) Dev Board**

Size: B Date: 2026-06-23

Rev: 1.0

KiCad E.D.A. 10.0.5

Id: 1/2

